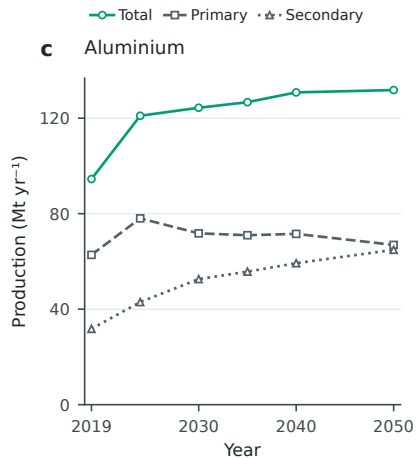
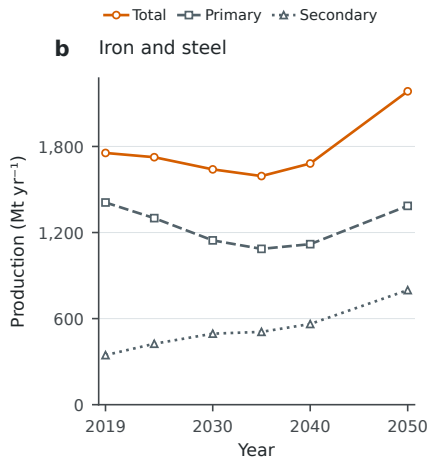
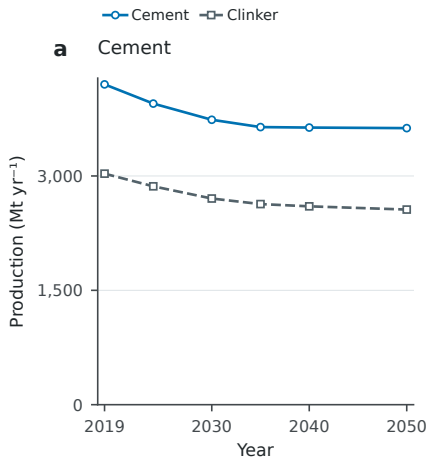


Material production and production routes

NDC · medium CE · EU | 2019-2050



Draft assumption: blank activity values = zero; reported prices and intensities retain gaps.

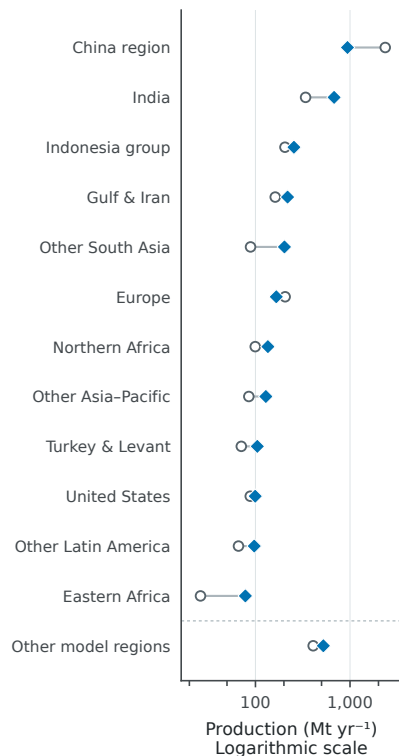
The geography of industrial production

NDC · medium CE · EU | 2019–2050

○ 2019 ◆ 2050 (sector colour)

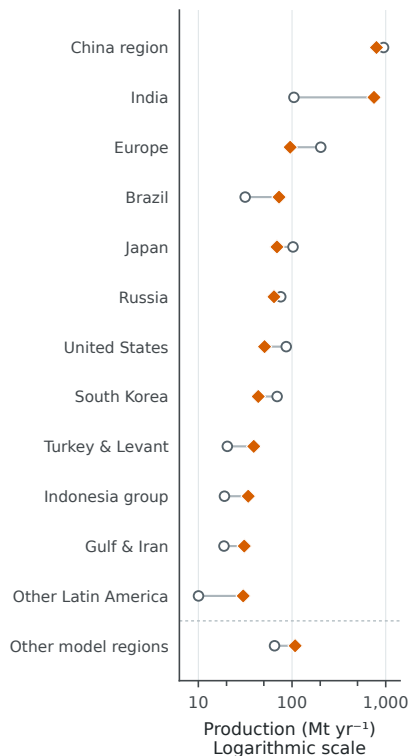
a Cement

Top 12: 85.6% of 2050 output



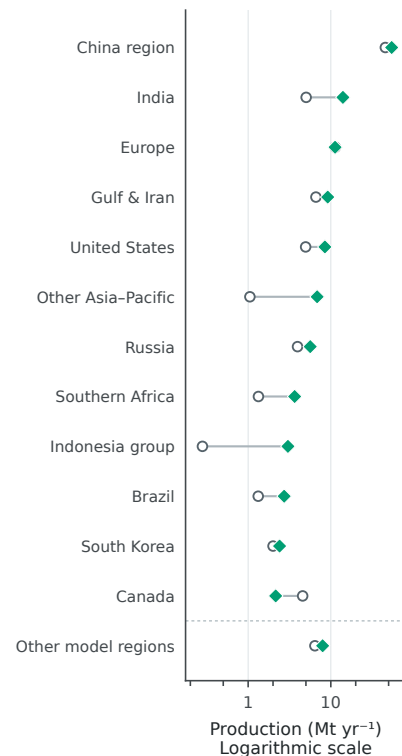
b Iron and steel

Top 12: 95.1% of 2050 output



c Aluminium

Top 12: 94.0% of 2050 output

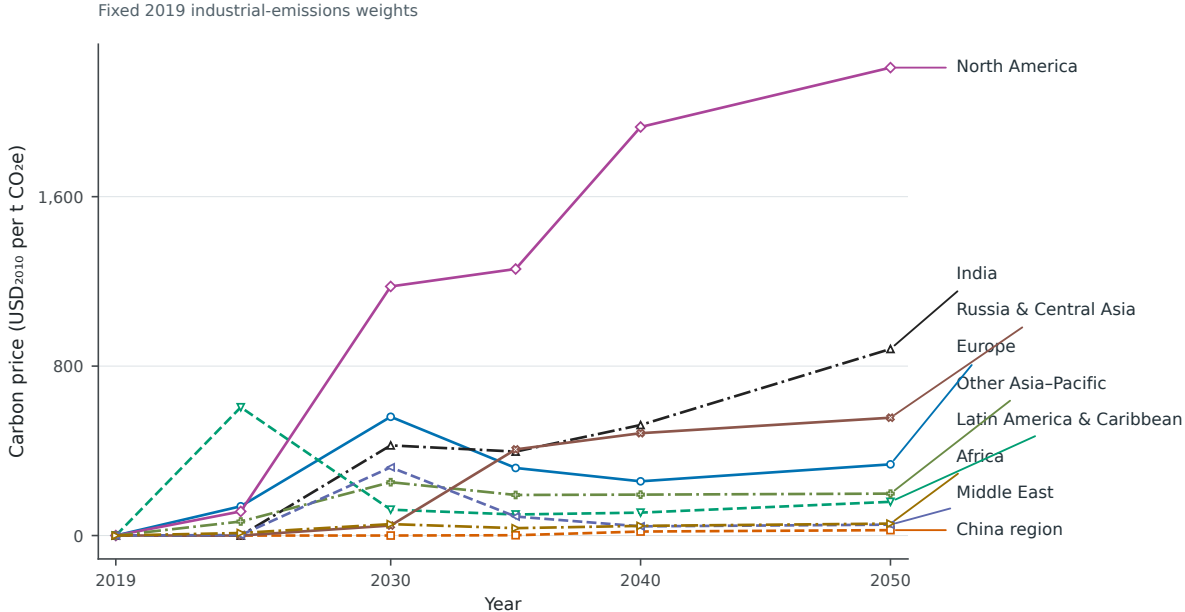


Europe combines five model regions. Indonesia group = Indonesia, Philippines and Viet Nam.

Draft assumption: blank activity values = zero; reported prices and intensities retain gaps.

Regional carbon-price indices

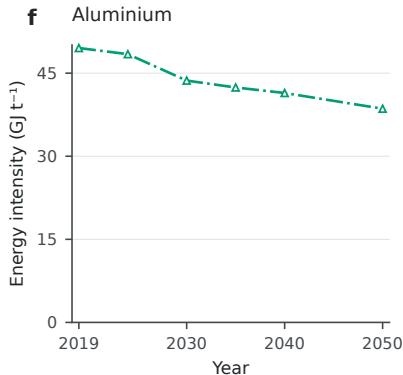
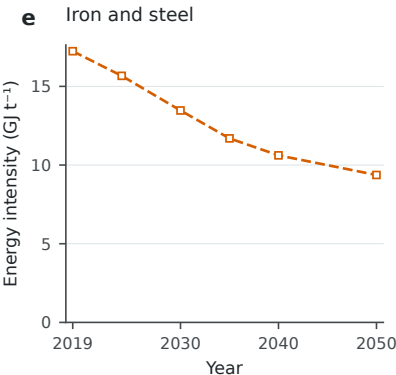
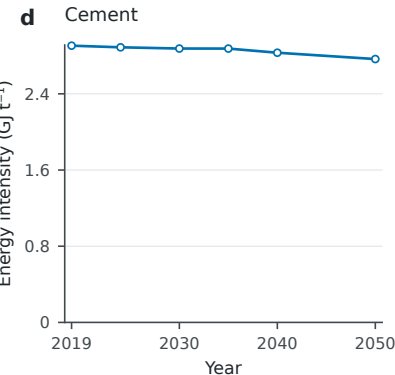
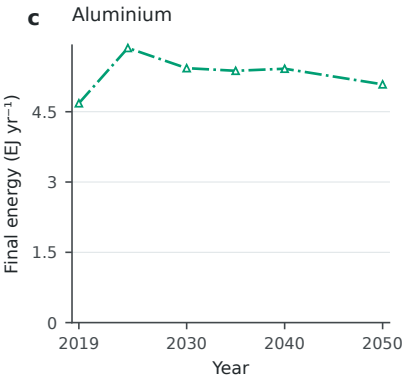
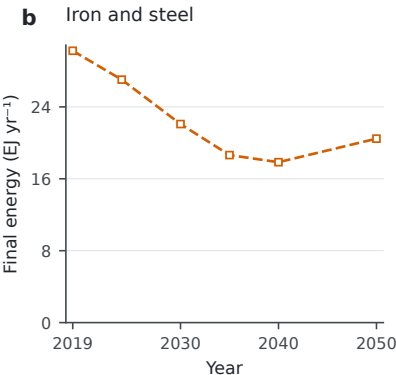
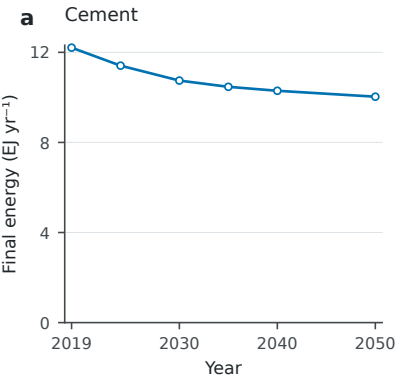
NDC · medium CE · EU | 2019–2050



Draft assumption: blank activity values = zero; reported prices and intensities retain gaps.

Industrial energy requirements

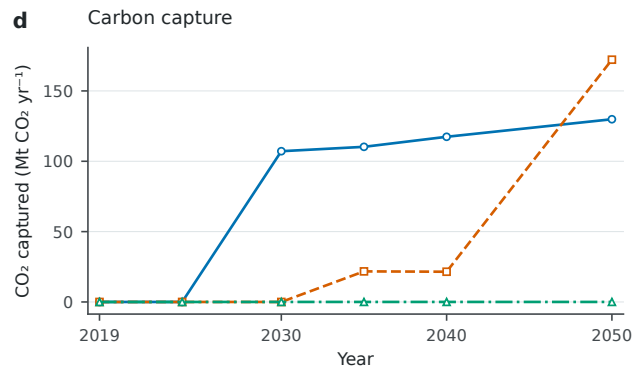
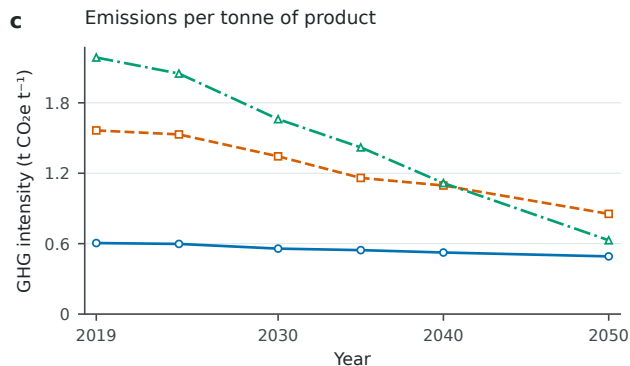
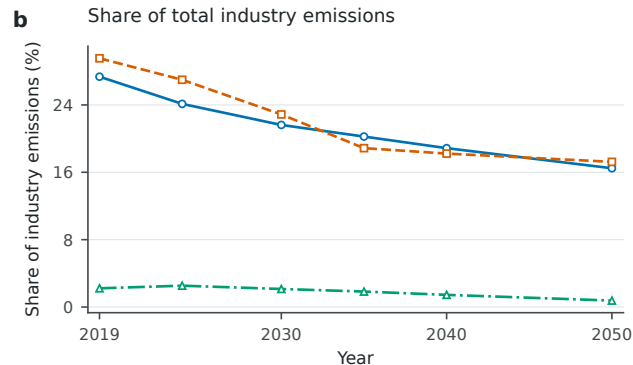
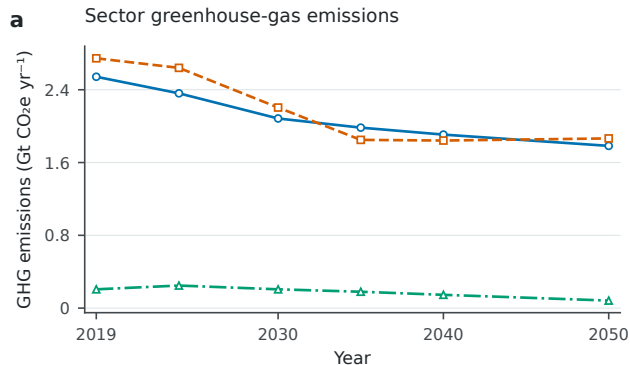
NDC · medium CE · EU | 2019-2050



Industrial emissions and carbon capture

NDC · medium CE · EU | 2019–2050

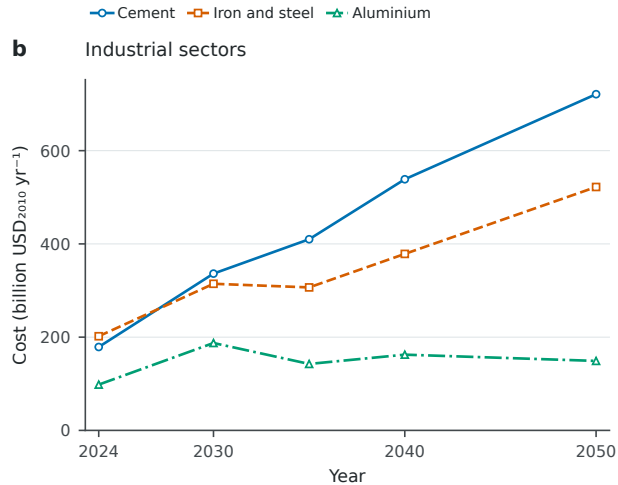
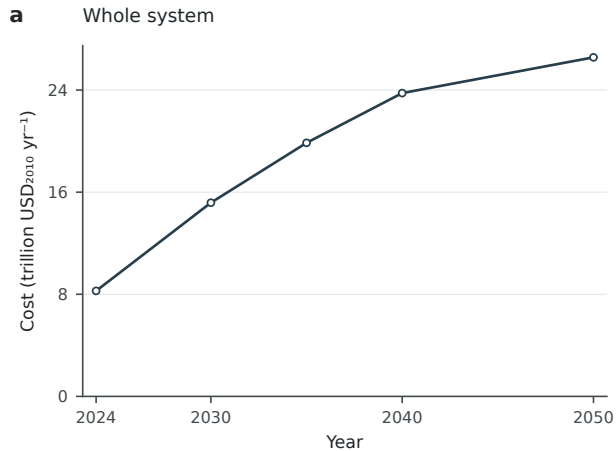
—○— Cement —□— Iron and steel —△— Aluminium



Draft assumption: blank activity values = zero; reported prices and intensities retain gaps.

Annualised costs, 2024-2050

NDC · medium CE · EU | 2024-2050



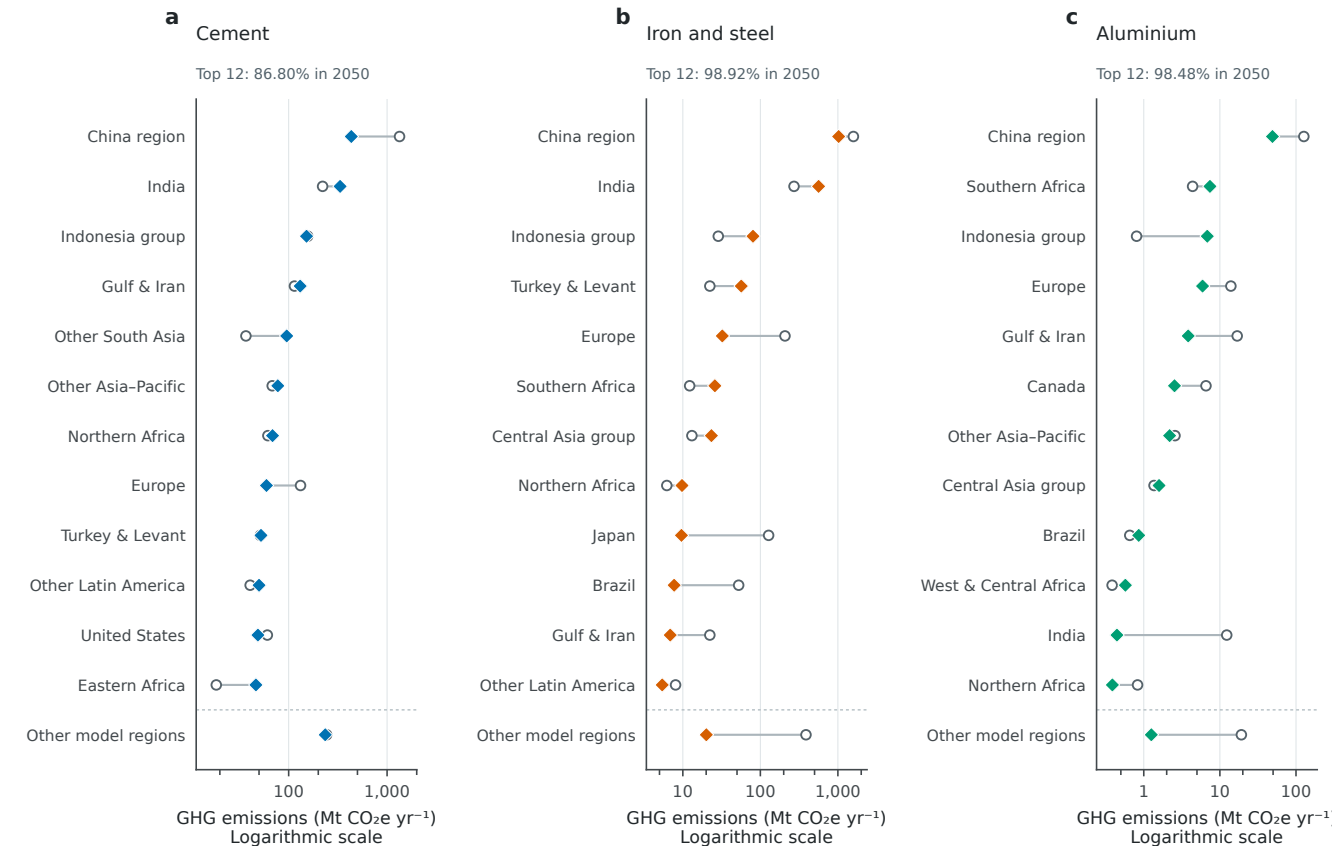
Main view starts in 2024 because of anomalous initial-year costs; the full period is retained in Figure S2.

Draft assumption: blank activity values = zero; reported prices and intensities retain gaps.

The geography of industrial emissions

NDC · medium CE · EU | 2019–2050

○ 2019 ◆ 2050 (sector colour)



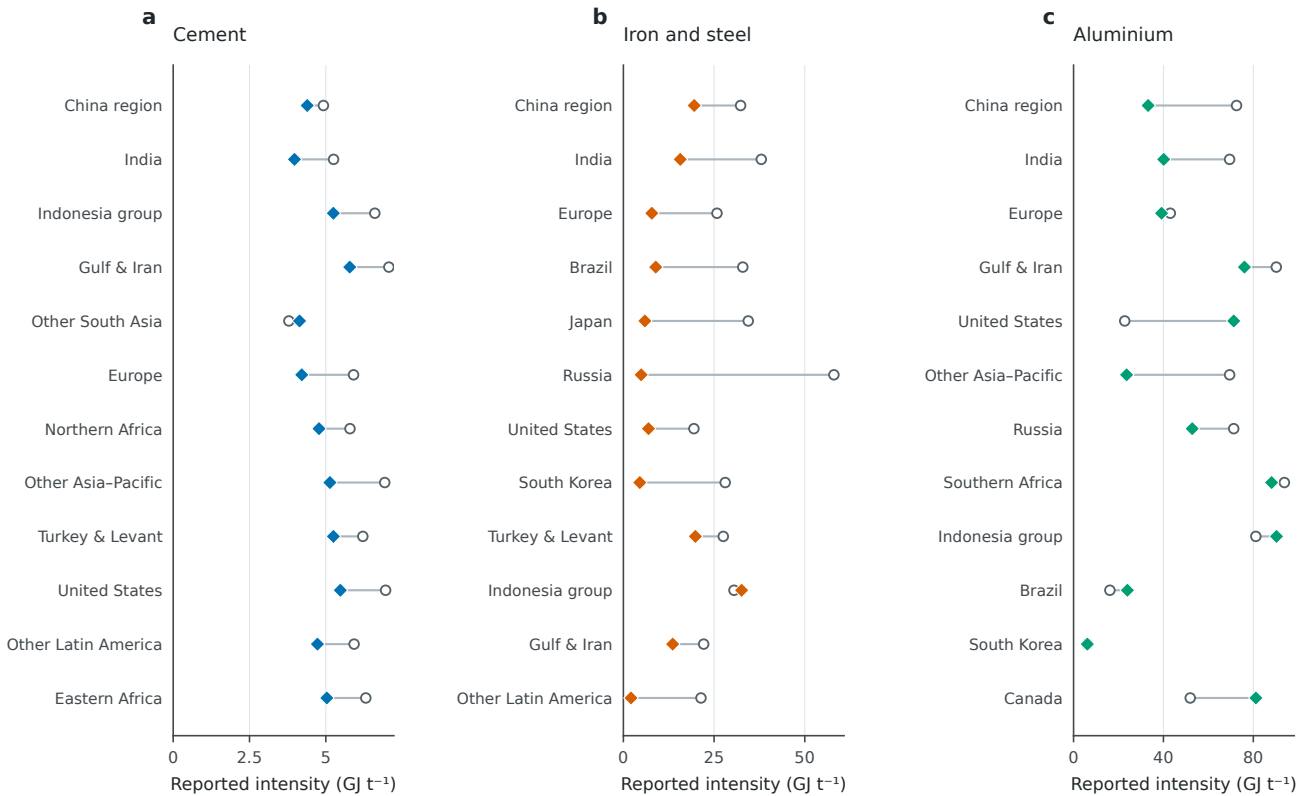
Europe combines five model regions. Indonesia group = Indonesia, Philippines and Viet Nam.

Draft assumption: blank activity values = zero; reported prices and intensities retain gaps.

Energy intensity in the leading producer regions

NDC · medium CE · EU | 2019–2050

○ 2019 ◆ 2050 (sector colour)

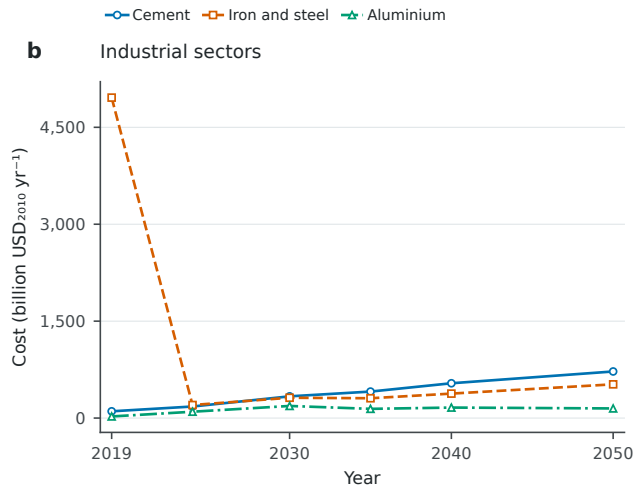
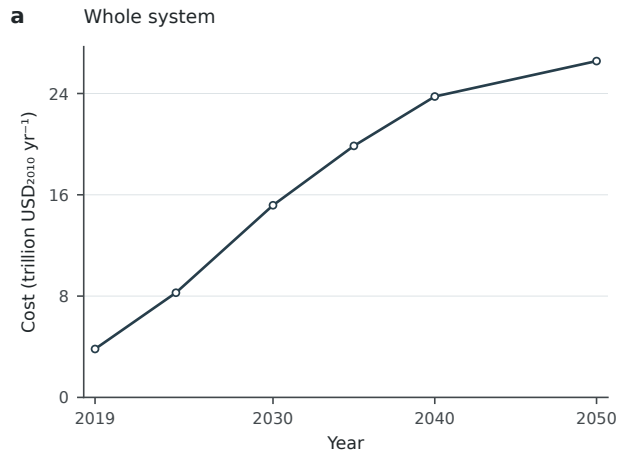


Europe combines five model regions. Indonesia group = Indonesia, Philippines and Viet Nam.

Draft assumption: blank activity values = zero; reported prices and intensities retain gaps.

Annualised costs: full source period

NDC · medium CE · EU | 2019–2050



Full source period shown; 2019 costs require checking before interpretation.

Draft assumption: blank activity values = zero; reported prices and intensities retain gaps.